Symbol	Parameter	Conditions		Min	Typ	Max	Unit
PGA BW	PGA bandwidth for different inverting gain	Gain = -3		-	GBW/4	-	MHz
		Gain = -7		-	GBW/8	-	
		Gain = -15		-	GBW/16	-	
en	Voltage noise density	at 1 KHz	output loaded with 4 kΩ	-	360	-	nV/√ Hz
		at 10 KHz		-	160	-	
I_{DDA}(OPAMP)	OPAMP consumption from V_{DDA}	Normal mode	no Load, quiescent mode, follower	-	1.6	2.8	mA
		High- speed mode		-	1.8	3	
		Normal mode	OPAMPint, follower	-	0.8	1.6	
		High- speed mode		-	1	1.8	

1. *R_{LOAD} is the resistive load connected to V_{SSA} or to V_{DDA}.*
2. *R2 is the internal resistance between the OPAMP output and the OPAMP inverting input. R1 is the internal resistance between the OPAMP inverting input and ground. The programmable gain amplifier (PGA) is calculated as 1 + R2/R1.*

5.3.22 Timer characteristics

The parameters given in Section 5.3.22, Table 62, and Section 5.3.22 are specified by design, not tested in production.

Refer to Table 47. I/O static characteristics for details on the input/output alternate function characteristics (output compare, input capture, external clock, PWM output).

Table 61. TIMx characteristics

Symbol	Parameter	Conditions	Min	Max	Unit(1)
$t_{res(TIM)}$	Timer resolution time	-	1	-	$t_{TIMxCLK}$
		$f_{TIMxCLK}$ = 144 MHz	6.9	-	ns
fEXT	Timer external clock frequency on CH1 to CH4	-	0	$f_{TIMxCLK}/2$	MHz
		$f_{TIMxCLK}$ = 144 MHz	0	72	
ResTIM	Timer resolution	TIMx (except TIM2/TIM5)	-	16	bit
		TIM2/TIM5	-	32	
$t_{COUNTER}$	16-bit counter clock period	-	1	65536	$t_{TIMxCLK}$
		$f_{TIMxCLK}$ = 144 MHz	0.007	455.1	µs
t_{MAX_COUNT}	Maximum possible count with 32-bit counter	-	-	4, 294, 967, 296	$t_{TIMxCLK}$
		$f_{TIMxCLK}$ = 144 MHz	-	29.826	s

1. *TIMx, is used as a general term in which x stands for 1, 2, 5, 6, 7, 8, 12, 15, 16, 17.*